Classification and Dimensionality Reduction using PCA

Lab Assignment 8 Report

Shrawanthi P. (Roll No: 26120098) Shreepriyadarshini RM. (Roll No: 26120099)

10th November 2026

1 Objective

In this lab, we test four classifiers using Python and NumPy:

- Gaussian Mixture Model (GMM)
- Linear Discriminant Analysis (LDA)
- K-Nearest Neighbours (KNN)
- Support Vector Machine (SVM)
- **Four Datasets:**
  - *Synthetic Data:* 20,000 points in 16 dimensions for 2 classes.
  - *MNIST Digits:* Grayscale handwritten images ($28 \times 28 = 784$ pixels) of digits 1, 8, and 9.
  - *Image Features:* 3,000 samples of 36 pre-computed features from 5 classes.
  - *TIDIGITS Audio:* 1,350 speech recordings of spoken digits 3, 7, and 9, represented by 39-D MFCC feature frames.

2 Workflow

1. **Split Data:** Split each dataset into train, validation, and test sets.
2. **Scale Features:** Normalised features with the mean and standard deviation of the training set: $z = (x - \mu)/\sigma$.
3. **Speech Setup:** We used two types of audio representations: (a) average all frames into one 78-D vector (mean and std), and (b) use all 39-D frames directly with a GMM.
4. **PCA Projection:** Obtain the top K directions with the highest variance in the training data, and map all data to K dimensions.
5. **Train and Test:** Trained and tested all four models on the original features and PCA features. Evaluate the performance using accuracy and confusion matrices.

3 Outputs, Results, and Observations

3.1 Synthetic Dataset (16-Dim vs. PCA 8-Dim)

Split into 12,000 train, 4,000 validation, 4,000 test points. The number of features that are reduced by PCA is 16. 8 dimensions were retained accounting for 72.7% of the variance.

Classifier	Orig. Val	Orig. Test	PCA Val	PCA Test	Change (Δ)
GMM	96.88%	96.43%	95.47%	95.25%	-1.18%
LDA	92.63%	92.85%	88.15%	88.15%	-4.70%
KNN	**99.40%**	**99.35%**	**96.90%**	**96.77%**	-2.58%
SVM	99.02%	98.78%	96.67%	96.65%	-2.13%

Table 1: Synthetic Dataset's Accuracies (16-Dim vs. PCA 8-Dim).

	Original Features (16-D)		PCA Features (8-D)	
True Class	**Pred. Class 0**	**Pred. Class 1**	**Pred. Class 0**	**Pred. Class 1**
Class 0	1961	15	1902	74
Class 1	11	2013	55	1969

Table 2: KNN confusion matrix result of 4,000 points of test.

Observations:

- If the classes have curved boundaries, then KNN (99.35%) and SVM (98.78%) are optimum. LDA is less accurate (92.85%) due to its ability to only draw straight lines.
- All models performed with less accuracy when PCA was applied.The largest accuracy drop(-4.70%) was shown by LDA because of the dropped dimensions useful informations were obtained to discriminate between the classes.

3.2 MNIST Handwritten Digits (784-D vs. PCA 40-D)

Digits 1, 8, and 9 have 21,660 in total, where the dataset was split into 14,833 train, 3,709 validation, and 3,118 test images.784 pixels were reduced to 40 features by using PCA. The difference in fittings was 56.6% of the variance that was retained from pixels.

Classifier	Orig. Val	Orig. Test	PCA Val	PCA Test	Change (Δ)
GMM	94.10%	93.94%	91.20%	90.92%	-3.02%
LDA	96.60%	96.54%	96.40%	96.34%	-0.20%
KNN	**98.40%**	**98.33%**	**98.30%**	**98.30%**	-0.03%
SVM	97.10%	96.99%	97.90%	**97.82%**	**+0.83%**

Table 3: Accuracies on MNIST digits 1, 8, and 9 (784-D vs. PCA 40-D).

True Digit	Pred. Digit 1	Pred. Digit 8	Pred. Digit 9	Recall
Digit 1	1129	5	1	99.47%
Digit 8	2	958	14	98.36%
Digit 9	8	23	978	96.93%

Table 4: KNN confusion matrix for 3118 test images with 40 PCA features.

Observations:

- The result of KNN is very precise on raw pixels (98.33%). The data was reduced by 95% using PCA, but KNN still achieves an accuracy of 98.30% and runs much faster.
- By removing noisy background pixels, PCA is able to help SVM slightly (from 96.99% to 97.82%).
- The major errors occur between digits 8 and 9, where both have loops so the model got confused. Digit 1 has almost no mistakes.

3.3 Image Features Dataset (36-Dim vs. PCA 12-Dim)

This dataset has 3,000 samples with 5 classes (2,000 train, 500 val, 500 test).The features reduced from 36 to 12 using PCA (69.4% variance kept).

Classifier	Orig. Val	Orig. Test	PCA Val	PCA Test	Change (Δ)
GMM	43.40%	44.20%	43.20%	47.80%	**+3.60%**
LDA	43.80%	43.20%	44.00%	44.20%	**+1.00%**
KNN	43.40%	41.60%	40.60%	41.80%	+0.20%
SVM	**50.40%**	**51.60%**	**50.40%**	**49.60%**	-2.00%

Table 5: Accuracies on Image Features (36-D vs. PCA 12-D).

True Class	Pred. 0	Pred. 1	Pred. 2	Pred. 3	Pred. 4	Recall
Class 0	21	22	19	18	20	21.0%
Class 1	0	82	4	10	4	82.0%
Class 2	5	17	44	19	15	44.0%
Class 3	10	6	12	59	13	59.0%
Class 4	12	20	6	10	52	52.0%

Table 6: SVM confusion matrix on 500 test samples on original features.

Observations:

- The 5 classes are highly overlapping, and the accuracies of all models are low (41%–51%). There was very little recall in Class 0 (21%).
- The correlation between features were removed by using PCA in order to improve accuracy of GMM,this helps to stabilize the training of GMM.(from 44.20% to 47.80%),

3.4 TIDIGITS Spoken Digits (Digits 3, 7, 9)

1,350 recordings (810 train, 270 val, 270 test).There were differences in comparison of 78-D summary vectors to the sequential 39-D frame GMM.

Representation	Classifier	Orig. Val	Orig. Test	PCA Val	PCA Test	Chang
78-D Summary	LDA	37.41%	37.78%	31.48%	31.11%	-6.6
78-D Summary	KNN	29.26%	28.89%	32.96%	32.96%	**+4.0**
78-D Summary	SVM	38.89%	39.26%	32.96%	32.59%	-6.6
Sequential Frames	**Frame GMM**	**99.26%**	**98.89%**	**99.26%**	**98.89%**	**0.0**

Table 7: Analysis of TIDIGITS (78-D summary vectors vs. Frame GMM)

	78-D Summary (SVM)			**Sequential Frames (Frame GMM)**		
True Digit	**Pred. 3**	**Pred. 7**	**Pred. 9**	**Pred. 3**	**Pred. 7**	**Pred. 9**
Digit 3	30	26	34	89	1	0
Digit 7	27	36	27	0	90	0
Digit 9	25	25	40	2	0	88

Table 8: Confusion matrices of 270 test audio files (90 for each digit)

Observations:

- Taking an average over time was not very successful for LDA, KNN, and SVM ($\approx$ 29%–39%), which is almost equal to random guessing (33.3%) since the sound order is lost.
- The highest result among all test methods is the accuracy of the frame-level GMM, which is 98.89% for both original frames and 15-D PCA frames. The number of test files was 270, with only 3 misclassifications. It is essential to maintain individual sound frames for speech.

4 Summary

- **Speech Needs Frames:** Even though speech features was classified correctly over short intervals, averaging those features reduced classification accuracy to approximately 35%. However, evaluating individual frames with GMM gave a accuracy nearly 99% accuracy.
- **Effects of PCA:** PCA works well when it removes noise (SVM on MNIST, GMM on image features). However, accuracy can be decreased when the dropped features contain meaningful information (Synthetic data).
- **Speed Gain:** On MNIST, PCA reduced the number of features from 784 to 40, significantly reducing the computation time for KNN with almost no change in accuracy (98.33% to 98.30%).
- **Linear vs. Non-linear:** KNN and SVM perform better than LDA when class boundaries are non-linear or curved.

3.5 Performance Visualizations

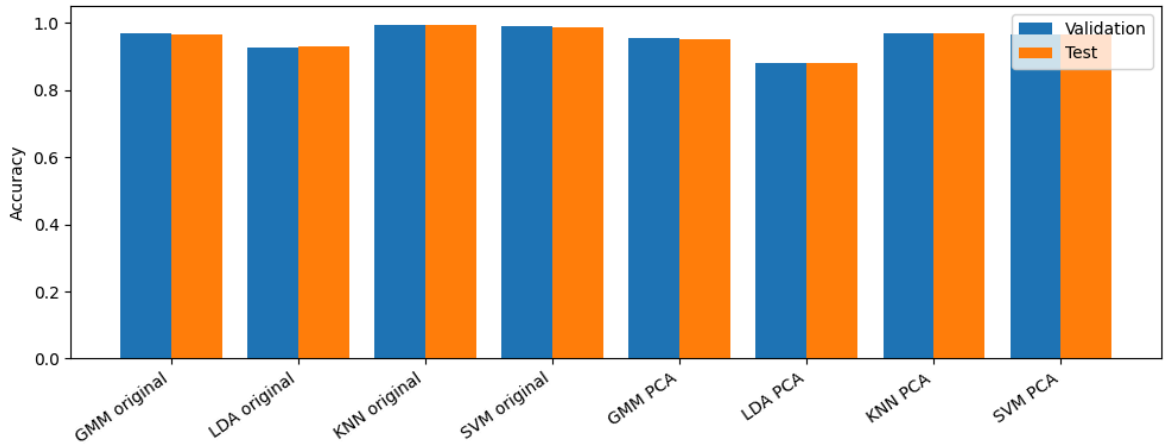


Figure 1: Synthetic Dataset (16-D vs. PCA 8-D) · Validation and Test Accuracies.

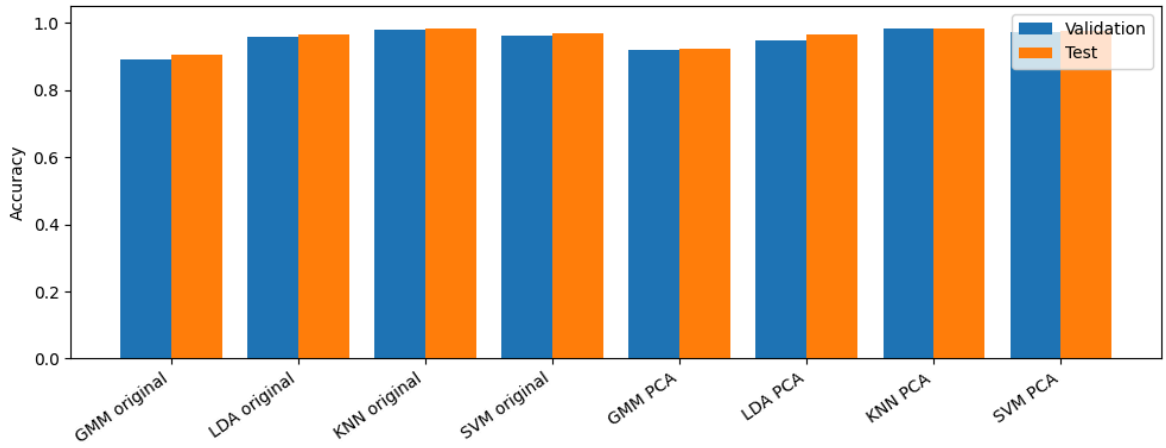


Figure 2: MNIST Handwritten Digits (784-D vs. PCA 40-D) · Accuracies and Execution Latency.

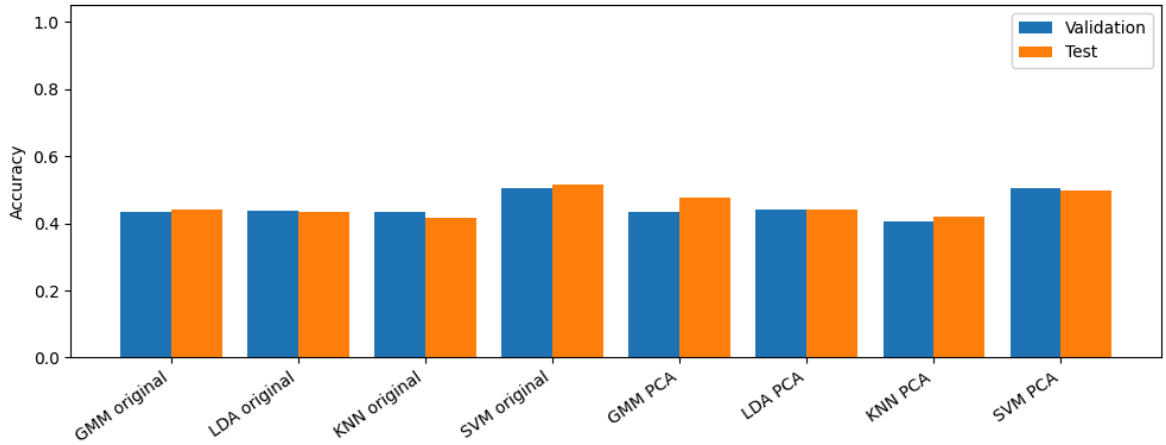


Figure 3: Image Features Dataset (36-D vs. PCA 12-D) · Classifier Accuracies (Val vs. Test).

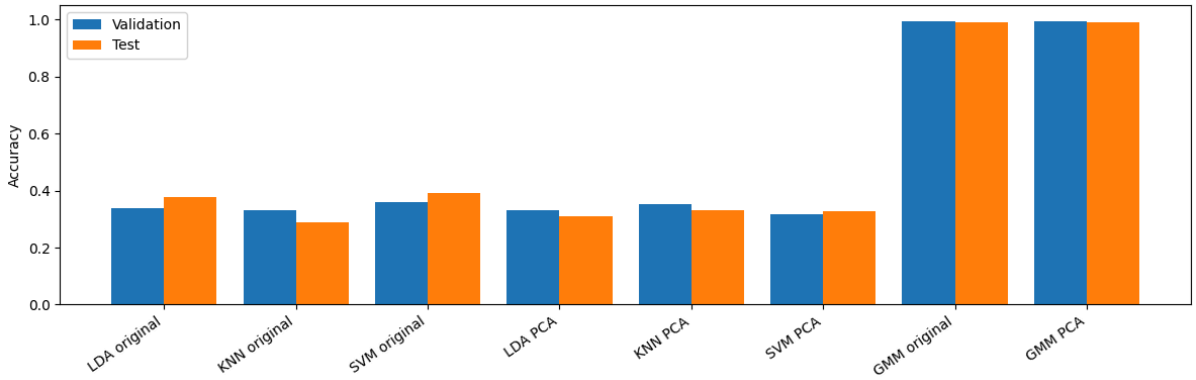


Figure 4: TIDIGITS Spoken Digits (Digits 3, 7, 9) · Summary Vectors vs. Sequential Frame GMM.